An Outcomes-Based Approach on Mathematics, Science and Technology
3 Hours Lecture

Module No.: 1
Topic: Assessing the Use of Technology in Math

<u>COURSE INTENDED LEARNING OUTCOMES</u>:

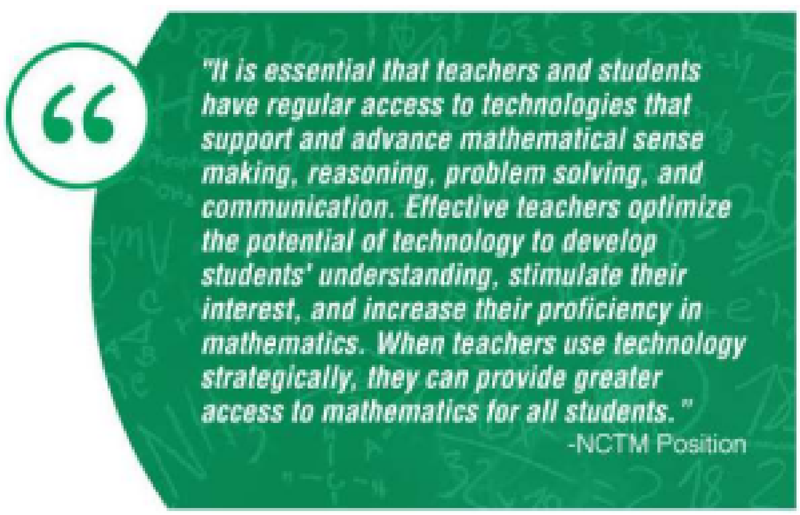
After reading this Module 1, you must be able to,

1. Recognize the capabilities of a variety of instructional technologies for use in Math Classroom.
2. Evaluate impacts of technology-enhanced lessons.

<u>TEACHING LEARNING ACTIVITIES</u>:

Technology provides dynamic opportunities for instruction in math classrooms. We can enhance the learning process and make concepts come alive through engaging and interactive media. We may also offer additional supports to address the needs of all learners and create customized learning experiences.

Research shows that teachers can integrate technology to help students grasp mathematical procedures and develop advanced mathematical proficiencies. The National Council of Teachers of Mathematics (NCTM) added that technological tools are <u>necessary for engaging students</u>.



8 Virtual Resources that Help Teach Mathematics

Here are some effective tools for teaching math with technology.

1. V<u>mathLive</u>

VmathLive features online mathematics competitions for students. This is a paid product.

The offering features four primary components:

a. An "Intelligent Teacher Dashboard" area that tracks student performance, including activity metrics, class progress for goals and milestones, and alerts for students who are struggling. b. A "Learn" component allowing students to progress at their own pace, through questions from 334 math topics.

c. A "Master" component that quizzes students on what they have learned. d. A "Play" component with real-time competitions from students across the United States.

2. Desmos

Desmos offers a free web-based graphing calculator as well as digital activities for grades 6-12.

The primary offering is a calculator that matches the functionality of many $100-plus TI calculators. It's a popular option for those who are looking to effectively teach math with technology. The other offering from Desmos, the digital activity center, presents several interactive lessons that students can perform on their own tablet or device. There is a section that enables teachers to create and share lessons with other teachers.

3. EquatIO

EquatIO is a tool for creating mathematical equations, formulas, Desmos graphs, and more on computers or Chromebooks.

The product seeks to digitize mathematics by simple input methods. Teachers can type, handwrite, or dictate expressions to add the result to a document, and there is a large library full of premade expressions. In other words, it's sort of a smart assistant that can replace pen and paper, as well as offer prediction capabilities. The product works with literacy software Read&Write (paid), which is helpful for blending accessibility and teaching math with technology.

4. Kahoot!

Kahoot! is an interactive game that presents multiple choice questions to students in the same classroom. This learning tool offers free and paid plans.

Teachers can search for premade questions and answers in a variety of subjects, or they can create their own (paid). Questions are projected on a screen in the classroom, and students select the answer on their devices. The highest scoring students are displayed before the next question, creating a fun, competitive atmosphere.

5. Online Games

Teaching math to students with online games is a popular method of engagement. Here are two free websites that feature a wide variety of interactive games for students.

a. Arcademics.com is an award-winning educational website that offers free multiplayer games, arcade math games, and more. The games adhere to Common Core math standards and span topics such as shapes, math functions, integers, fractions, and algebra.

b. Math Playground offers several math and logic games, as well as arcade-based activities and math videos. A small number of games require a paid subscription.

6. SMART Board Activities/Games

Classrooms equipped with SMART boards can take advantage of free activities and games available across the internet.

Sites such as SMART Exchange and iSmartboard.com help teachers find thousands of fun lessons, and they are organized by grade level. Many of the activities and games are submitted by teachers who are already using them for teaching math in the classroom.

7. Sumdog

Sumdog is an adaptive learning tool for teaching math with technology to students in grades K-8. Paid subscription plans unlock premium teacher tools.

Sumdog features game-based learning that motivates and engages students. On computers and tablets, students answer questions at their own pace, with activities geared towards their skill level. Positive reinforcement rewards students' achievements in the game, and teachers receive assessment data to guide lessons and examine individual student needs. Effectiveness studies and case studies have found improvements in student achievement with the product.

8. YouTube Videos

YouTube is a great source for interactive videos that help students learn mathematical concepts and practice what they've learned. Here are some of the most noteworthy YouTube channels for math instruction:

a. Khan Academy has a multitude of free videos covering basic and advanced concepts. b. Mathademics offers short videos on elementary and middle school topics, focusing on subjects such as geometry and fractions.

c. Numberphile features colorful videos on topics such as Pi, prime numbers, and games like Yahtzee.

Teachers can also create their own YouTube videos, for a unique perspective on teaching math with technology. Examples range from a more elaborate and wacky show to a simpler and more focused clip in the real world.

Impacts of technology-enhanced lessons.

The processes of teaching and learning have experienced significant changes in the 21st century due to changes taking place in the society. One of the changes has been with regard to the application of technology in learning. Pierce and Ball (2009) acknowledge that teachers today have an extensive range of sophisticated technology available to them.

Technologies such as Mathematics software, scientific calculators, spreadsheets, and statistical packages have become common place in many classrooms. This is in sharp contrast to the

traditional mathematics classroom, which was dominated by pen and paper. BECTA (2003) asserts that the technology has "changed the nature of teaching and learning in maths" (p.1). The changes have been widespread due to government action through the National Curriculum. BECTA (2003) states that the National Curriculum has made the use of ICT in Mathematics a statutory requirement and all Secondary school students therefore have to make use of technology in their studies.

Positive Impacts

1. Use of technology serves as a motivation for teachers due to the positive outcomes achieved.
2. Technology fosters the development of a culture of effective teaching by teachers. 3. Teaching mathematics with technology enhances the teacher's ability to teach students about problem solving.
4. Technology has been used to help cope with some of the "hard to teach" aspects of secondary mathematics.
5. Teaching mathematics with technology gives the teacher more tools with which to offer instructions to the students.

Negative Impacts

1. Using technology might reduce the control that the teacher has in the class setting. 2. The use of technology in teaching mathematics requires major changes to teaching practices. 3. The increased workload for the teacher as he/she learns how to use the new technology 4. Teaching efficiency for some teachers will reduce as they learn how to incorporate technology in their teaching practice.

ASSESSMENT TASK #1

Instruction: Answer the following questions.

1. What are the 8 Virtual Resources that Help Teach Mathematics.
2. Enumerate some positive and negative impacts of using technology in teaching. Give at least 2 each.
3. In your own opinion, does technology-enhanced lessons really effective in the learning process? Why?

REFERENCE:

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